Anchoring Global Efficiency: Visakhapatnam Leads India's Entry into the World's Top 20 Container Ports

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'Visakhapatnam is a very special port city of India with an extremely rich tradition of trade and business. Visakhapatnam, being an important port in ancient India, was part of the trade route to West Asia and Rome thousands of years ago, and it still remains the central point of India's trade in today's day and age.'

Honourable Prime Minister Narendra Modi, Visakhapatnam, 12 November 2022 (Press Information Bureau [PIB] 2022)

On 19 June 2024, the Ministry of Ports, Shipping and Waterways shared news that stirred pride across India's maritime sector. The World Bank and S&P Global had released the 2023 Container Port Performance Index, and for the first time, nine Indian ports were among the world's top 100 (World Bank and S&P Global 2024). One port in particular stood out—not only for being the highest ranked in India but also for the rapid progress it had made. Speaking on the occasion, Union Minister of Ports, Shipping and Waterways, Sarbananda Sonowal, announced, "*Visakhapatnam Port made it to the top 20 ports of the world at 19 in 2023, recording a marked improvement from 115 in 2022 – another first for India*"(PIB 2024).

As congratulatory messages flowed in from across the country, M. Angamuthu, chairperson of the Visakhapatnam Port Authority (VPA), reflected on the year-long effort that had begun when he took charge in May 2023. He could not help but think of every team member who had believed this transformation was possible. This was not just a leap in rankings—it was a story of people, purpose, and partnership. Its success underscored a broader lesson for public infrastructure and public sector undertakings: performance transformation requires not only capital investment but also leadership, technology adoption, and cross-team and policy coherence working in tandem. As one VPA official put it, "*The ranking is not a goal to validate our performance or that of any Indian port – it is the outcome of deliberate, long-overdue reforms and initiatives that directly addressed challenges the port had grappled with for decades.*"

Visakhapatnam Port Authority: From Heritage to Innovation

Visakhapatnam Port, also known as Vizag Port, began operations on 19 December 1933. Established by the Bengal Nagpur Railway to facilitate mineral exports, primarily manganese ore, the port initially operated with just three berths and handled about 0.13 million tonnes of cargo annually. Over the decades, it expanded to 31 berths, and handled over 82.62 million tonnes of cargo in fiscal year 2024–2025 (VPA Newsletter 2025a).

Located on India's eastern seaboard near the Strait of Malacca and supported by strong rail and road connectivity, Vizag Port historically served as a vital gateway for trade with the vast hinterlands[1] of Indian states such as Andhra Pradesh, Telangana, Odisha, Chhattisgarh, and Maharashtra. It also serves as a second gateway for Nepal's export–import trade (VPA n.d.).

The port's evolution had been guided by the VPA, a statutory body formed under the Major Port Trusts Act of 1963 and reconstituted under the Major Port Authorities Act of 2021 and the rules published thereunder. This reconstitution granted greater functional autonomy to the VPA Board[2], enabling more agile decision-making and fostering collaboration with the private sector (Major Port Authorities Act 2021).

VPA's mission was to be a major partner in the logistics supply chain on India's east coast, aligning with national initiatives like the Maritime Amrit Kaal Vision 2047[3], Sagarmala[4], and PM Gati Shakti[5] (Exhibit 1; Ministry of Ports, Shipping and Waterways 2015, 2021, 2023; Ministry of Commerce & Industry, n.d.). The authority oversaw marine operations, cargo handling, rail logistics, environmental safeguards, and the integration of smart port technologies.

In 2024, Vizag Port made headlines by securing the 19th position in the Container Port Performance Index (CPPI)—a data-driven global benchmark for port efficiency (see Exhibit 2).

The CPPI 2023 used both administrative and statistical methods to measure port efficiency, focusing on how long ships spent in port. Key indicators included the speed at which ships were berthed, the efficiency of cranes, and the duration of containers' stay in the port. Vizag Port's performance beat global averages on these parameters (World Bank and S&P Global 2024) (Exhibit 3). Indeed, Vizag outperformed many larger, better-funded global ports. But behind this achievement lies a story of tough choices, legacy hurdles, and quiet groundwork that began long before the recognition.

A Port at the Crossroads: Challenges Confronting Angamuthu at Vizag Port

In May 2023, M. Angamuthu, an Indian Administrative Services officer, assumed charge as VPA chairperson. He brought with him extensive experience from his previous role as chair of the

[1] Hinterland refers to the inland area served by a port, from which goods are brought for export and to which imported goods are distributed.

[2] As per the Major Port Authorities Act of 2021, the Board of Major Ports is a permanent body comprising the chairperson and deputy chairperson of the port; one member each from (i) the state government in which the Major Port is located, (ii) the Ministry of Railways, (iii) the Ministry of Defence, and (iv) the Customs Department; one member nominated by the central government, ex officio; and two members representing the interests of the employees of the major port authority.

[3] A comprehensive road map published by the Ministry of Ports, Shipping and Waterways outlines the transformation of the Indian maritime sector over the next 25 years, with the goal of making India a leading maritime nation by 2047

[4] The Sagarmala Programme is a flagship initiative of the Ministry of Ports, Shipping and Waterways aimed at promoting port-led development in India.

[5] PM Gati Shakti is a digital platform run by the Government of India to bring 16 ministries, including railways, roadways, shipping, and waterways together for integrated planning and coordinated implementation of multimodal infrastructure connectivity projects.

Agricultural and Processed Food Products Export Development Authority.[6] At the time of his joining, the port had already undergone significant infrastructure development initiated by his predecessors, including the expansion and modernization of the container terminal, widening of port roads and national highway, among others. The new chairperson had to build on this momentum, with an urgent need to mechanize and digitize port operations to enhance efficiency and grow business, amid rising competition from private ports in a post–COVID-19 volatile trade environment.

Globally, container cargo[7] makes up only about 20% of total cargo by volume, but nearly 70% of total cargo value (World Bank and S&P Global 2024). In simple terms, any port aiming to move up the value chain and contribute more to the country's economic growth must focus on container handling. Yet at Vizag, the port's cargo remained dominated by bulk cargo, such as iron ore, coal, and bauxite, shipped to countries like the People's Republic of China or routed to domestic steel and power plants in southern India. This was because eastern ports like Vizag served a hinterland rich in minerals but lacked the high-value, processed-goods industries that drive container cargo growth. This stood in contrast to ports in Gujarat and Maharashtra, which were supported by strong industrial bases and dense manufacturing clusters.

This disconnect became acute post–COVID-19, as global disruptions tripled container charges and worsened shortages. Exporters—many handling time-sensitive goods such as seafood and farm produce—found themselves unable to secure timely berths without paying premium fees. "Why pay more just to get timely service?" they asked. Hence, the VPA needed to undertake a major overhaul to increase its container cargo business while balancing its established bulk cargo operations.

To further understand the on-ground issues, the new chairperson talked to local exporters, private partners, shipping and customs agents, other public sector undertakings, and stakeholders. *"From the joining day itself, I did not mind meeting 50 people a day and just listening. Most of the solutions have come out from my stakeholders only,"* said Angamuthu. The more he listened, the more the layers of challenge became clear.

Institutional Rigidities and Legacy Inefficiencies

Vizag Port was, in many ways, a textbook example of a legacy government institution. Until the early 2000s, and before the implementation of the Major Ports Authority Act, 2021, it was entirely government-run—managed by boards with a large number of trustees and centralized control by the federal government. This limited the port's legal ability to make independent decisions regarding operations, private partnerships, and tariff setting. Even basic commercial discretion— such as offering a discount to attract more container traffic—was subject to scrutiny. *"If I subsidize a rate,"* one senior official explained, *"the audit will question whether I am showing undue favour. We are not a private enterprise—we are a government board."* This fear of audit overreach bred inertia and made risk-taking virtually impossible.

Infrastructure and Connectivity Bottlenecks

Even as the port worked to address internal limitations, its location posed unique logistical challenges. Many ports are built on isolated coastlines, but the Vizag Port is embedded within a growing urban city. Unlike ports that benefit from dedicated freight corridors or exclusive port roads, trucks carrying cargo must navigate congested city streets, competing with local traffic for space. The local population frequently complained about excessive noise and dust emissions caused by the movement of heavy

[6] An apex organization of the Ministry of Commerce and Industry, Government of India, created specifically for export promotion of agriculture and allied products from India.
[7] Container cargo involves goods packed into standardized shipping containers, offering flexibility and security, while bulk cargo refers to unpackaged goods, often raw materials, shipped in large quantities

container trucks. Vizag lacked even basic infrastructure, such as flyovers or separate access lanes for port-bound vehicles. A senior container freight station operator summed it up bluntly: *"We have been asking for 50 acres just for truck parking and empty container yards. It is basic, but it is not there."* Poor last-mile connectivity increased turnaround times, pushed up costs, and eroded VPA's competitiveness.

Competitive Pressure from Private Ports

Meanwhile, private ports, especially those operated by large groups, were offering exporters and shipping lines exactly what the VPA could not: quick turnaround, responsive management, seamless paperwork, and flexible pricing. With their modern terminals and customer-friendly policies, these ports began siphoning off Vizag's container traffic. Stakeholders were candid in their warnings: "If big, private players like Adani (one of India's largest multinational conglomerates) buy and conquer, all government ports will be out of the container game."

A Fragmented, Low-Value Cargo Base

Adding to the complexity was the nature of Vizag's cargo mix. Unlike ports such as Chennai or Mundra, which cater to industrial clusters exporting electronics, automobiles, and high-end machinery, Vizag dealt primarily in low-margin goods such as paper, seafood, and steel billets. These cargoes could not absorb high logistics costs. The absence of local manufacturing or major special economic zones (SEZs) meant limited outbound container loads. Most SEZs in the region were oriented toward intermediate goods rather than value-added exports. Without a dense industrial ecosystem, container trade remained sparse and fragmented.

Process Delays in Customs and Cargo Handling

Even as physical infrastructure improved, procedural bottlenecks remained. Customs processes that were critical for timely cargo clearance, were stuck in outdated protocols. Modern practices like Direct Port Delivery and Direct Port Entry, which allow trusted clients to move goods directly to or from the port, bypassing intermediate handling at container freight stations, were underutilized. The classification of "trusted clients" was restrictive, and shipments often required cumbersome "stuffing and de-stuffing" at container freight stations, especially for Less-than-Container Load cargo.[8] For exporters, these delays translated into missed shipping windows and rising costs.

A Tipping Point

"It was a question of surviving the business with competitiveness," reminisced Angamuthu. It was not just about cranes and berths. It was about shifting mindsets, digitizing processes, attracting new industries, and proving that a government-run port could match the agility of its private-sector rivals. For the VPA, the challenge was clear: how to accelerate the transformation of a legacy bulk port into a modern, container-focused gateway, overcoming institutional inertia and stakeholder scepticism, while ensuring Vizag could compete in a rapidly evolving logistics landscape.

Implementation Process: Operational Excellence Rooted in Measurable Outcomes

Before 2022, container operations at Vizag Port were limited in scale. As trade volumes increased and shipping lines demanded quicker turnaround, it became clear that the existing setup would not be enough. The need for faster, more reliable operations brought both VPA and its private partner,

[8] Less than container load refers to a shipping method where multiple smaller shipments are combined into a single container for transportation. This is a cost-effective option for businesses with shipments that do not require a full container. LCL allows for sharing of transportation costs and optimizes space utilization.

Visakha Container Terminal Private Limited (VCTPL), to the same table. They understood that operational excellence required greater synergy—where VPA provided policy and infrastructure support, and VCTPL delivered on-the-ground execution and technology upgrades. Both sides recognized their roles in shaping a more capable and competitive port.

Enhancements Done by VCTPL

Operating since 2003 under the Design-Build-Finance-Operate-Transfer Model (DBFOT),[9] the VCTPL invested in transforming the container terminal's infrastructure, improving container cargo handling and supporting service delivery. For them, investing heavily in Vizag Port meant clear long- term gains. More capacity meant more containers and larger ships, which brought in more revenue.

In December 2021, a major transformation began at the VCTPL. The quay[10] at Terminal 1 was extended from 450 m to a massive 895 m, making it the longest continuous quay on India's east coast. This expansion enabled larger ships to dock simultaneously, boosting throughput capacity. With a draft[11]of 16.5 m—deepest in the country—Visakha Container Terminal can now accommodate container vessels up to 390 m in length overall, meaning ships nearly four football fields long can berth safely.

But lengthening the quay was only part of the story. Three super post panamax rail mounted quay cranes[12] were also added. Each of these cranes has an outreach of 65 m, allowing them to reach across a ship's width, and a safe working load of 65 tons, enabling the efficient lifting of heavy containers. The addition of these cranes increased the total number of quay cranes at the port from four to seven, significantly improving cargo-handling speed. At the same time, the number of rubber-tired gantry cranes nearly doubled, with nine electric units fitted with GPS-based position detection, in addition to 10 electric ones at Terminal 1. In addition to this, five new electric reach stackers were added in the fleet replacing diesel reach stackers. These upgrades meant containers could be moved faster within the yard, improving overall workflow and reducing congestion, while reducing port's carbon footprint of about 80 tons annually.

The container storage yard[13] was upgraded with an additional 41 acres, enabling 2,600 more ground slots and raising the daily handling capacity to 18,000 TEUs, thereby further reducing the congestion and improving yard productivity. To facilitate seamless cargo evacuation, a new six-lane kiosk-enabled gate complex was commissioned in August 2022, alongside the existing three-lane facility, streamlining trailer movement and reducing turnaround time for cargo vehicles.

To support the increasing volume of refrigerated cargo, Terminal 2 added 300 new reefer plug points with 24/7 diesel generator backup, complementing the existing 350 plug points at Terminal 1. These upgrades strengthened the port's capacity to handle India's expanding marine export trade.

Together, these infrastructure improvements—particularly the use of advanced twin-lift crane technology, which enables lifting of two containers simultaneously, and the extension of berths—helped reduce vessel turnaround times significantly. Ships could now dock, unload, and depart faster than before, increasing the port's throughput and productivity.

[9] The Design-Build-Finance-Operate-Transfer (DBFOT) model is a public–private partnership (PPP) structure where a private entity is responsible for designing, building, financing, and operating a project, with the project being transferred back to the public sector at the end of the concession period.
[10] The docking area where ships unload cargo
[11] Draft is the minimum water depth needed for a ship to safely berth.
[12] These cranes are towering machines on rails that load and unload containers from ships.
[13] A container storage yard is a designated area within a port or terminal where containers are stored before they are loaded onto a ship or released after being unloaded.

VPA's Operational Drive

Operational efficiency at Visakhapatnam Port was not achieved through infrastructure upgrades alone. A series of ecosystem-level interventions, both within the container terminals and across the wider port area[14] and ecosystem, enabled smoother, end-to-end coordination. The VPA board agreed to bring bold reforms in pricing. Premium charges for priority berthing were scrapped by August 2023. This move was strategic: while it involved a short-term revenue dip, the objective was to attract greater cargo volumes in the long run.

The Custom Bound Area[15] was expanded from 160 acres to 339 acres in 2024, allowing better cargo flow management across terminals. Supporting these operational changes, the team analysed over a decade of the port's performance data—tracking cargo trends, shipping line behaviour, and trade flows. This deep dive generated valuable insights for targeted improvements. By benchmarking against leading global ports, Visakhapatnam was able to align its performance indicators with international standards, ensuring every decision was grounded in robust, comparative data.

These efforts culminated in VCTPL's investment in the adoption of NAVIS N4, a fully integrated digital platform and the latest generation Terminal Operating System that unified all port stakeholders—shipping lines, customs, importers, and exporters—through a single-window interface. By streamlining documentation, enhancing real-time visibility, and reducing the human interface in customs compliance, this digital transformation not only boosted operational efficiency but also fostered greater transparency, accountability, and faster service delivery.

This required the adoption of new technology-driven processes by port users. Hence, VPA and VCTPL shared frequent circulars, conducted multiple training and awareness programs (including shipping and customs agents, truck drivers) to demonstrate new digital processes (including NAVIS N4 and the RFID gate system). This helped address any grievances or issues faced by stakeholders and incorporate their feedback. Further, VPA took measures to strengthen its communication and relation with various port stakeholders.

Empowering Public–Private Partnership Operators

The partnership with private players strengthened after the Major Port Authority Act, 2021, gave VPA's board the legal freedom to act. As the chairperson explained, *"Private players do require flexibility as they come with their own autonomy, processes, and specialized skills."* With this, the chairperson initiated the practice of conducting monthly meetings with each public–private partnership (PPP) operator[16] (separately), along with all heads of departments of the port to review and understand their work and requirements at one table. To attract more business, the port increased its trade meets, stakeholder conclaves, and outreach with industry players across India. These engagements resulted in assurances from participating container liners to increase their cargo volumes through Vizag Port, helping build robust industry partnerships.

During this period, Visakhapatnam Port also began transitioning to a landlord model. Under this model, the VPA became responsible for providing all necessary infrastructure, particularly facilities beyond the port gates, such as roads, truck parking stations, and railway connectivity. For VCTPL specifically, a designated four-lane road was constructed in 2023 from the city's busy common junction road to the container terminal to ensure the continuous movement of containers. As the owner and

[14] The Vizag Port spans over 7,500 acres of Vizag's landmass.
[15] A custom-bound area, often referred to as a bonded warehouse, is a secure storage facility within a customs area where imported goods can be stored without immediate payment of customs duties. These areas are designed for goods that are either still in transit or awaiting processing and export.
[16] Presently, the VPA has eight PPP operators with two major partners: JM Baxi Ports & Logistics running VCTPL for container cargo, and Vedanta Limited running the Vizag General Cargo Berth Private Limited for bulk cargo

regulator of the land and assets, VPA focused on strategic oversight, while private companies managed the daily operations. This shift marked a significant change in port management, as it separated regulatory oversight from day-to-day activities. It also created opportunities for private operators to invest directly in infrastructure upgrades and technology.

Introducing Additional Incentives with a Win-Win Approach

To encourage volume growth and reward long-term commitment, the VPA introduced measures to support transshipment activities with a discount of up to 80% for the transshipment containers to boost movement. As a result, operating costs decreased for both the port and its partners. To further boost cargo volumes, the board approved a policy offering financial concessions to stakeholders who increased their cargo volumes beyond their own 3-year historical average. This "incremental cargo" scheme was designed to recognize and incentivize consistent performance rather than one-off transactions. Under the scheme, logistics providers, shipping lines, and other cargo handlers who demonstrated growth above their previous average were eligible for rebates on port charges. The more cargo they brought in over their baseline, the greater the concession they received.

This approach marked a shift from short-term, transactional engagements to a more strategic focus on sustained partnerships. It created a clear and predictable incentive structure that motivated stakeholders to expand business through the port.

Creating a Seamless Multimodal Ecosystem and Stakeholder Coordination

Coordination within the Port Working Committee—which included the VPA, VCTPL, exporters, customs clearing and forwarding agents, the Vizag Stevedores Association,[17] the Vizag Steamship Agents[18] and other port users—was strengthened. Regular meetings, supported by clear action reports and a dedicated WhatsApp channel, enabled swift issue resolution and ensured accountability

In addition to this, the VPA and VCTPL deepened engagement with Customs, CONCOR, Indian Railways, and more than 65 container lines to improve coordination and efficiency. Plans for a 100- acre logistics hub and cold storage facilities were proposed, and a new truck parking terminal was inaugurated. These measures ensured smooth cargo movement and prevented congestion at the container terminal.

Strengthening Internal Monitoring and Strategic Planning

Angamuthu constituted a unique Port Infrastructure Committee to break down silos and strengthen cross-departmental coordination. The committee—comprising heads of department, deputy heads, and senior port officials—was organized into five-member teams to conduct monthly, on-site inspections across port zones, including visual documentation of any improvements or upgrades required in these areas. Every month, the general administration department consolidates the reports from all teams, and a review meeting is held under the chairpersonship of Angamuthu. A senior official from the civil department commented: *"This monthly meeting keeps us on our toes. The teams have become more proactive and competitive in their work. The best part is that members with desk jobs get to visit the port operations on site. This was not possible earlier due to their nature of work. Now, the sanctions and approvals of various projects over e-files are faster as departments are better coordinated and understand the various port operations more logically."*

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[17] A stevedores association is an organization that represents the interests of companies or individuals involved in the loading and unloading of cargo from ships

[18] Steamship agents, also known as shipping agents or port agents, are individuals or companies that represent shipping lines or vessel owners in a specific port or territory.

In parallel, the chairperson mandated each department to craft annual action plans with clear targets and business strategies, integrating feedback and suggestions from both internal teams and external stakeholders. This collaborative process began each January and culminated by late February or March with an annual stakeholders' meet, where the chairperson personally unveiled the unified action plan and vision for the year ahead.

According to Angamuthu, there was no resistance from staff or port users to the changes. He said, *"Whatever we were doing was not just business for the port; it was incremental business with human perspective where everyone felt included in the decision-making process."* He added, *"They were our 'partners in progress.' Everyone benefited, and that was why we face no resistance."*

Impact

Between FY 2022–2023 and FY 2024–2025, Vizag Port demonstrated measurable progress across multiple performance indicators. Cargo throughput increased from 73.75 million tons in FY 2022–2023 to 81.09 million tons in FY 2023–2024, and further to a record 82.62 million tons in FY 2024–2025, indicating consistent growth in trade volumes and port-handling efficiency (VPA 2025a). This was accompanied by a sharp rise in container traffic, which grew from 0.52 million TEUs in FY 2022–2023 to 0.67 million TEUs in FY 2023–2024, reflecting the port's growing appeal to shipping lines and exporters (VPA 2024). Vizag also contributed around 314,199 metric tonnes of marine and aqua cargo in FY 2023–2024—about one-third of India's total in this segment— highlighting its sector importance (MPEDA 2024). In parallel, private partner VCTPL, operating under the landlord port model, invested ₹720 crore across four new terminal projects aimed at automating operations and enhancing handling capacity, underscoring the transformative role of PPPs in the port's evolution (VPA 2024).

From Eastern Shores to Global Score: Way Ahead

No port operates in a vacuum. From customs clearance to rail logistics, from cold chain to container tracking—each piece of the system matters. Vizag treated port efficiency as a shared outcome, not an individual metric. Going ahead, VPA aimed to accelerate the infrastructure upgrades: the port's highway was being expanded from four to ten lanes under Bharatmala,[19] with four dedicated lanes for port traffic. Flyovers were under design to bypass 11 railway crossings[20] in and around the city, thereby decreasing traffic congestion, in collaboration with the Rail Vikas Nigam Limited, a PSU under the Ministry of Railways. Stockyards[21] were being redeveloped in a phased manner, with proper drainage systems, hardscaping of open cargo areas, and improved access roads to cargo zones and railway sidings, in partnership with MECON,[22] to ensure safe, efficient cargo storage— even during monsoon season. This would further enhance operational efficiency at the port. Further, the VPA aimed to transform into a landlord model port by 2030 with 100% cargo handling by PPP terminals. This sent a strong signal to other major ports in India, and aligned closely with the Maritime India Vision 2030, which aimed to move the majority cargo at major ports through similar PPP models.

[19] Bharatmala Pariyojana is a major road development project in India, approved in 2017, that is focused on improving connectivity and reducing logistics costs.

[20] Railway network at Port of Visakhapatnam is the largest among Indian ports with over 200 km rail length, over 30 sidings.

[21] A stockyard at a port is a designated area, often open-air, used for storing and managing cargo, particularly bulk materials, before or after these are loaded onto or unloaded from ships. These yards are essential for efficient port operations, allowing for the accumulation of goods, blending, and staging for onward transportation

[22] MECON Limited, formerly known as Metallurgical & Engineering Consultants (India) Limited, is a central public sector undertaking under the Ministry of Steel, Government of India

However, Angamuthu also raised a critical concern—underutilization. The VPA was still operating at only 60% capacity, mainly held back by competitive pressures. Many nearby private ports were performing even below this level. Despite this, new ports were under development, raising a larger question for the VPA and VCTPL: Are they building faster than they are scaling demand?

At this point, Angamuthu advocated for a regional port coordination authority that spanned 400–500 km of coastline, encompassing major, minor, private, and state ports. *"We compete for the same cargo. But sometimes I can handle cargo X better, and my neighbouring port is better for cargo Y. Why not share infrastructure, optimize capacity, and build a win-win model?"* said Angamuthu.

In his opinion, a regional authority could manage vessel rerouting, berth scheduling, and shared evacuation logistics. This cooperative model could drastically reduce anchorage delays and logistics costs—the key friction points in global shipping.

Sudeep Banerjee, Senior Vice President and Terminal Head of the VCTPL, highlighted the urgent need to anchor port operations in a stronger local and regional cargo base. He suggested strengthening the hinterland and logistics base. *"A port should be fed by its own hinterland. Vizag is still dependent on far-off states. That is risky and unsustainable,"* explained Captain Sudeep Banerjee.

Lessons Learned

Three key lessons stood out from Vizag's journey

First, institutional transformation was rooted in collaborative partnership. VPA's turnaround demonstrated how strategic public–private collaboration—when backed by clear roles, trust, and accountability—could deliver operational excellence at scale. The outcome of the synergy between the VPA and VCTPL under the PPP was visible:

- VCTPL's quay expansion, digitization, and adoption of NAVIS N4 helped reduce port call duration to 19.1 hours and berth idle time to under 1.5 hours, far exceeding global averages.
- Vizag became India's largest marine and aquaculture cargo hub, handling nearly 30% of the country's seafood exports in 2023–2024.
- The port had been attracting some of the world's largest vessels like MSC Monica Christina and later a 15,486 TEU Trans-Pacific mothership in June 2025

As affirmed by Venugopal M. Harbour Master, VPA: *"Despite being nearly a day-and-a-half off the main international shipping lanes like the Colombo–Singapore corridor, vessels are consistently calling Vizag Port. The preference proves that companies trust the quality of services and the operational efficiency that we offer."*

Indeed, Vizag's location on the Upper Bay of Bengal could position it as a natural transshipment hub for the eastern subcontinent, including ports such as Kolkata, Haldia, and even extending to Myanmar. As argued by Sudeep Banerjee, this required greater investment in supporting infrastructure such as large-scale logistics parks, cold storage networks, and multi-user distribution centres that could catalyse trade, especially in value-added cargo like fruits and electronics— cargo that currently bypasses Vizag for Chennai due to superior ecosystem support. *"If we create the critical mass through enhanced rail links and transshipment planning, more mother vessels will come. Vizag can become northern India's preferred eastern gateway,"* noted Banerjee.

Second, policy autonomy must be matched with agile and inclusive leadership. The VPA board leveraged the autonomy provided by the Major Ports Authority Act, 2021 to recognize bottlenecks and act decisively on infrastructure, incentives, and revenue models that enabled rapid modernization. The chairperson institutionalized monthly inspections, multi-stakeholder planning, grievance redressal channels, and transparent performance reporting. As a senior VPA official said, *"Our leadership wants to hear everybody and promotes an open communication with port users. You explore our website. You can see every big or small information. You can access the existing and ongoing port infrastructure projects (with timelines and completion rate), port performance data, business plan, and development initiatives—all in one place."* This shows that empowered boards and clear legal mandates must be activated by participatory leadership that intrinsically motivates the teams to excel toward shared progress.

Third, modern ports must balance operational growth with environmental responsibility and a citizen-centric approach. In line with Harit Sagar Green Port Guidelines, 2023,[23] sustainability became a central theme of Vizag Port's transformation:

- Over 650,000 trees were planted on 650 acres of port land, with a million more pledged across Vizag city by 2028.
- Landscaping and illumination were done around port roads, and a public promenade road is being developed leading up to International Cruise Terminal to provide a close-up view of vessels coming into the port.

With this, the VPA aimed to turn port spaces into shared civic zones and foster a sense of shared ownership among local residents, thereby reducing community resistance. For the VPA, strengthening public support is a crucial factor for port-led development. Angamuthu explained this vision, *"The port spans over 7,618 acres of the Vizag city. We aim to change the citizens' perception that the port is a pollution-emitting industrial area. Today, we have families doing morning and evening walks alongside the landscaped roads, clicking pictures at selfie-points, etc. People should feel part of the port ecosystem with pride, especially my own employees and partners. After all, before anything else, they are the citizens of Vizag."*

Vizag has redefined its role—not just as a port but as a driver of "ease of business" and "ease of living" for the entire region.

This shift from a busy port to an ecosystem enabler is the next chapter. As Angamuthu said, *"It does not matter whether it is a private or government port. As long as we are building the economy together, it is the right direction."*

Exhibits

Exhibit 1: Alignment with Maritime Amrit Kaal Vision 2047 (MAKV 2047)

The *Maritime Amrit Kaal 2047*, unveiled by the Honorable Prime Minister in 2023 at the Global Maritime India Summit, aimed to position India as a global maritime leader by 2047, coinciding with 100 years of independence. It built on the *Maritime India Vision 2030*, with objectives that included developing world-class ports, promoting inland water transport, coastal shipping, and sustainability—supported by an estimated investment of Rs80 trillion over 25 years. Key goals included achieving carbon neutrality at major ports, promoting domestic hydrogen production, developing 25 cruise terminals, and making India one of the world's top five shipbuilding nations.

VPA's efforts to achieve global excellence in port operations aligned with MAKV 2047 vision of making India a maritime leader. The port's rise to the 19th place in the Container Port Performance Index (CPPI)—with metrics such as 27.5 moves per crane hour and a 21.4-hour vessel turnaround time—reflected operational efficiency, a key vision objective. Sustainability initiatives, such as planting 1 million trees, supported the vision's focus on carbon neutrality, while the development of the Vizag International Cruise Terminal aligned with its emphasis on promoting cruise tourism, as mentioned in the vision. VPA's strategy to handle more cargo, aiming for 90 million tonnes in 2025–2026, contributed to the vision's economic growth targets.

Alignment with Sagarmala

Sagarmala, launched in 2015 by the Government of India, was a flagship program to promote port- led development by harnessing India's 7,500 km coastline and 14,500 km of navigable waterways. Its objectives included modernizing ports, improving connectivity, encouraging port-led industrialization, developing coastal communities, and reducing logistics costs for export–import and domestic trade, with an investment of Rs5.54 lakh crore planned from 2015 to 2035. VPA explicitly coordinated with Sagarmala, as stated by the chairperson, with investments close to ₹3,769 crore in various projects under the program. These efforts supported port-led industrialization and coastal community development, core Sagarmala pillars.

Alignment with PM Gati Shakti

VPA's mission to enhance multimodal connectivity aligned with PM Gati Shakti's integrated planning approach. The port was improving rail connectivity, with 60% of container cargo moving by rail and road infrastructure, including a 12 km four-lane link to National Highway-16, ensuring seamless cargo movement. These efforts reduced logistics costs and enhanced competitiveness, aligning with PM Gati Shakti's goals of integrating infrastructure schemes such as Sagarmala and improving last-mile connectivity.

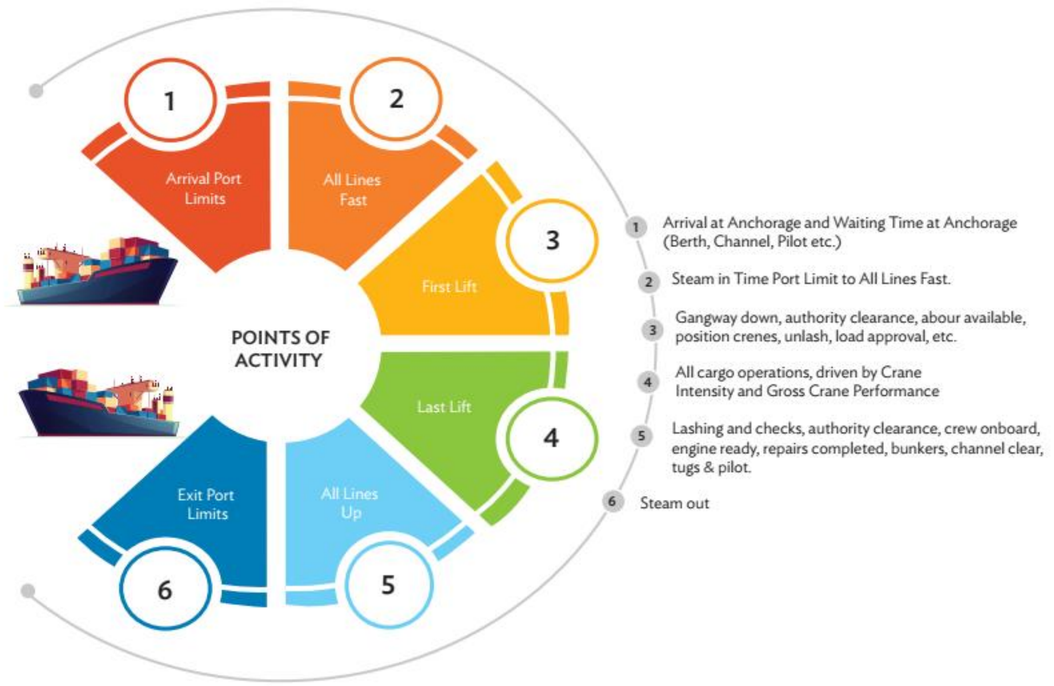
Source: Provided by VPA.

Exhibit 2: The CPPI Ranking Explained

Since 2009, the World Bank and S&P Global have published the *Container Port Performance Index (CPPI)*—a rigorous global benchmark designed to assess how efficiently container ports operate. Unlike perception-based rankings, the index is built entirely on empirical data, aimed at supporting efficiency improvements in container port operations and the optimization of port calls. The dataset includes move counts for each individual port call undertaken by 10 of the world's largest liner shipping companies, which collectively operate nearly 80% of global fleet capacity. The CPPI highlights which ports perform well and identifies where vessels are delayed, infrastructure lags, and operations demonstrate speed and precision (World Bank and S&P Global 2024).

In 2023, performance timestamp data were captured for 194,198 port calls involving 253.7 million container moves at 876 container terminals in 508 ports worldwide (World Bank and S&P Global 2024). These metrics provided insights into both operational speed and coordination across port systems—both critical factors influencing international trade costs and overall supply chain efficiency. Because no surveys or subjective assessments were used, the index offered a neutral, data-driven view of performance. It tracked six stages of a ship's visit—from arrival to departure—using Automated Identification System data and operational timestamps. Every container ship port call was broken down into six distinct steps. which are illustrated in Figure 1. "Total port hours" was defined as the total time elapsed between a ship's arrival at a port (whether at port limits, pilot station, or anchorage zone—whichever event occurs first) and its departure from the berth after completing cargo exchange. The time from berth departure (All Lines Up) to leaving the port limits was excluded because any port performance loss related to departure delays—such as pilot or tug availability, mooring readiness, channel access and water depth, forecast completion time, communication, and ship readiness—would occur while the ship is still alongside the berth. Additional time resulting from these causes would, therefore, be captured during the period between the Last Lift and All Lines Up (berth departure).

Exhibit 3: Anatomy of a Port Call



Note: The CPPI 2023 measured the performance for Stages 2–5.
Source: World Bank CPPI 2023 Report.

Exhibit 4: Vizag Port's Performance vs Global Averages

Indicator	Description	CPPI Global Average (2023)	VPA/VCTPL Performance (FY 2023–2024)
Total port call duration	The period a vessel spends at a port, from its arrival to its departure	40.5 hours	19.1 hours
Berth idle time	The period a berth remained unutilized while a vessel was docked	3.7 hours	<1.5 hours
Gross crane productivity	The number of container moves handled by a crane per hour	23.5 moves/hour	27.9 moves/hour
Pre-berthing delay	The duration between a vessel's arrival at the port, waiting at anchorage or designated waiting area, and when it is secured at the berth	~5–6 hours typical	1.9 hours
Dwell time	The period a container spends within a port before being either gated out (import) or loaded onto a vessel (export)	4 days for the United Arab Emirates and South Africa 7 days for the United States and 10 days for Germany	2.6 days

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Study Questions

1. What were the most significant operational changes that enabled Vizag Port's dramatic improvement in global rankings?
2. What are the key elements that made the VPA–VCTPL partnership effective? If you were advising another public port, what lessons from Vizag's PPP approach would you recommend replicating—and what would you do differently?
3. How can lessons from Vizag be applied to other Indian ports or inland logistics hubs?
4. How did Dr. Angamuthu's leadership style and stakeholder-centric approach drive change at Vizag Port?
5. How could Vizag Port institutionalize its recent gains—ensuring that the culture of stakeholder engagement, data-driven decision-making, and operational excellence would endure beyond the tenure of any one leader?

Amrit Gyaan Kosh is a dynamic repository of knowledge resources that captures and curates best practices in governance in form of case studies. This initiative is designed to reinforce the principles of *Atmanirbharta*, support the objectives of *Mission Karmayogi*, and contribute to the Hon'ble Prime Minister's vision of a citizen-centric, transparent, and accountable governance framework. These cases are published on iGOT, a learning platform for Government officials. For any query related to Amrit Gyaan Kosh, please write to gyaankoshcbc@gmail.com.

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